

SSILP Math

Beginner

ALGEBRAIC AND RATIONAL EXPRESSIONS
EQUATIONS



Session Outline

2

1. Algebraic expressions

2 Rational expressions

3. Equations

Learning objectives

- Learn how to Add, Subtract, Multiply Algebraic Expressions
- Use Special Product Formulas
- Use Factoring in expressions
- Rationalize denominator or numerator
- Solve linear, quadratic and other types of equations

[Simplifying using factoring formulas \(Khan academy\)](#)

[Multiplying rational expressions \(Organic chemistry tutor\)](#)

[Solving quadratic equations \(Organic chemistry tutor\)](#)

[Completing the square \(Organic chemistry tutor\)](#)

[Absolute value equations \(Organic chemistry tutor\)](#)

Beginner level videos for algebra

Skip if you already know this material

1. Algebraic expressions

Algebraic expressions

Overview

**Adding and
subtracting
polynomials**

**Multiplying
Algebraic
expressions**

**Factoring/
Expanding**

Special product formulas

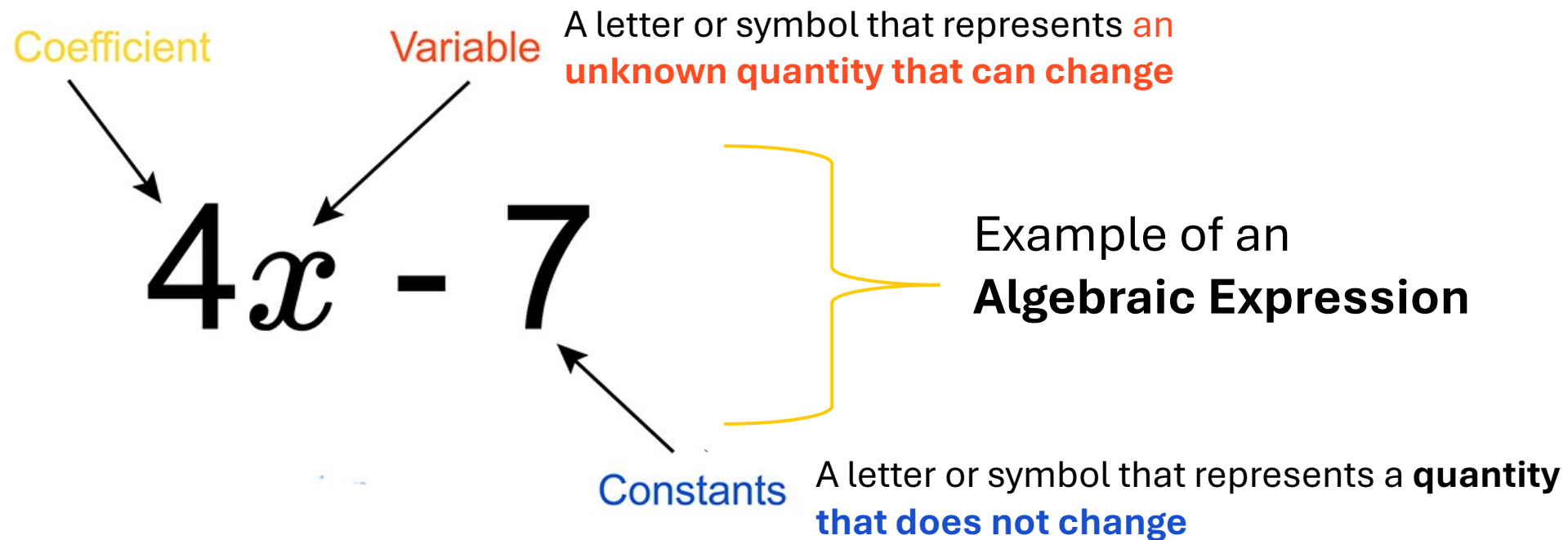
Factoring by grouping

Textbook Reference

Pre-calculus : Mathematics for Calculus
By Stewart, Redlin, Watson (7th Ed)

Chapter 1, Section 1.3:
Algebraic Expressions
Pg 25-33

What are the parts of an algebraic expression?



Arithmetic equations simply contain numbers and arithmetic operations,

an algebraic expression will always include one or more **variables**, as well as some numbers

Algebra has real world applications in Architecture and Design!

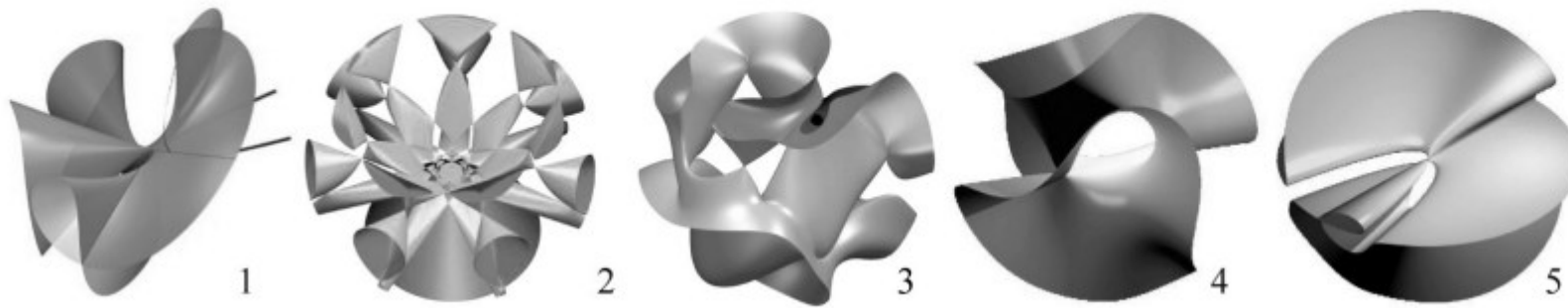


Figure 1
Algebraic surfaces by Oliver Labs (1,2), Herwig Hauser (4,5) and an anonymous creator



Figure 5
Le Corbusier's and Xenakis' Philips Pavilion (1,2), Ito's Guangzhou (3), Snohetta's Tubaloon (4)

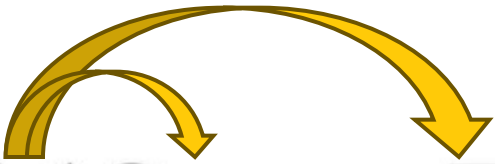
Exploration of the manifold novel shapes found in **algebraic geometry** and their application in architectural design

Polynomial and degrees of polynomials

Polynomial	Type	Terms	Degree
$2x^2 - 3x + 4$	trinomial	$2x^2, -3x, 4$	2
$x^8 + 5x$	binomial	$x^8, 5x$	8
$8 - x + x^2 - \frac{1}{2}x^3$	four terms	$-\frac{1}{2}x^3, x^2, -x, 3$	3
$5x + 1$	binomial	$5x, 1$	1
$9x^5$	monomial	$9x^5$	5
6	monomial	6	0

Distributive property for algebra


$$a(b + c) = ab + ac$$


$$4(2x + 3)$$

$$4 \cdot 2x + 4 \cdot 3$$

$$8x + 12$$

Adding and subtracting polynomials

(a) Find the sum $(x^3 - 6x^2 + 2x + 4) + (x^3 + 5x^2 - 7x)$.

(b) Find the difference $(x^3 - 6x^2 + 2x + 4) - (x^3 + 5x^2 - 7x)$.

SOLUTION

$$(a) \quad (x^3 - 6x^2 + 2x + 4) + (x^3 + 5x^2 - 7x)$$

$$= (x^3 + x^3) + (-6x^2 + 5x^2) + (2x - 7x) + 4$$

Group like terms

$$= 2x^3 - x^2 - 5x + 4$$

Combine like terms

$$(b) \quad (x^3 - 6x^2 + 2x + 4) - (x^3 + 5x^2 - 7x)$$

$$= x^3 - 6x^2 + 2x + 4 - x^3 - 5x^2 + 7x$$

Distributive Property

$$= (x^3 - x^3) + (-6x^2 - 5x^2) + (2x + 7x) + 4$$

Group like terms

$$= -11x^2 + 9x + 4$$

Combine like terms

Multiplying algebraic expressions

$$(a + b)(c + d) = ac + ad + bc + bd$$

$\uparrow$ $\uparrow$ $\uparrow$ $\uparrow$
F **O** **I** **L**

$$(2x + 1)(3x - 5) = 6x^2 - 10x + 3x - 5$$

$\uparrow$ $\uparrow$ $\uparrow$ $\uparrow$
F **O** **I** **L**

$$= 6x^2 - 7x - 5$$

Distributive Property

Combine like terms

Special Product formulas

SPECIAL PRODUCT FORMULAS

If A and B are any real numbers or algebraic expressions, then

1. $(A + B)(A - B) = A^2 - B^2$ Sum and difference of same terms
2. $(A + B)^2 = A^2 + 2AB + B^2$ Square of a sum
3. $(A - B)^2 = A^2 - 2AB + B^2$ Square of a difference
4. $(A + B)^3 = A^3 + 3A^2B + 3AB^2 + B^3$ Cube of a sum
5. $(A - B)^3 = A^3 - 3A^2B + 3AB^2 - B^3$ Cube of a difference

$$(x^2 + y^3)^2 = (x^2)^2 + 2(x^2)(y^3) + (y^3)^2$$

$$(A + B)^2 = A^2 + 2AB + B^2$$

Using special product formulas

Use the Special Product Formulas to find each product.

(a) $(3x + 5)^2$ (b) $(x^2 - 2)^3$

SOLUTION

(a) Substituting $A = 3x$ and $B = 5$ in Product Formula 2, we get

$$(3x + 5)^2 = (3x)^2 + 2(3x)(5) + 5^2 = 9x^2 + 30x + 25$$

(b) Substituting $A = x^2$ and $B = 2$ in Product Formula 5, we get

$$\begin{aligned}(x^2 - 2)^3 &= (x^2)^3 - 3(x^2)^2(2) + 3(x^2)(2)^2 - 2^3 \\ &= x^6 - 6x^4 + 12x^2 - 8\end{aligned}$$

Factoring common factors

$$a(b + c) = ab + ac$$

$$3x^2 + 6x$$

$$3 \cdot x \cdot x + 2 \cdot 3 \cdot x$$

$$3x(x + 2)$$

find the factors that are
common to **both terms**

 **FACTORING** 

$$x^2 - 4 = (x - 2)(x + 2)$$

 **EXPANDING** 

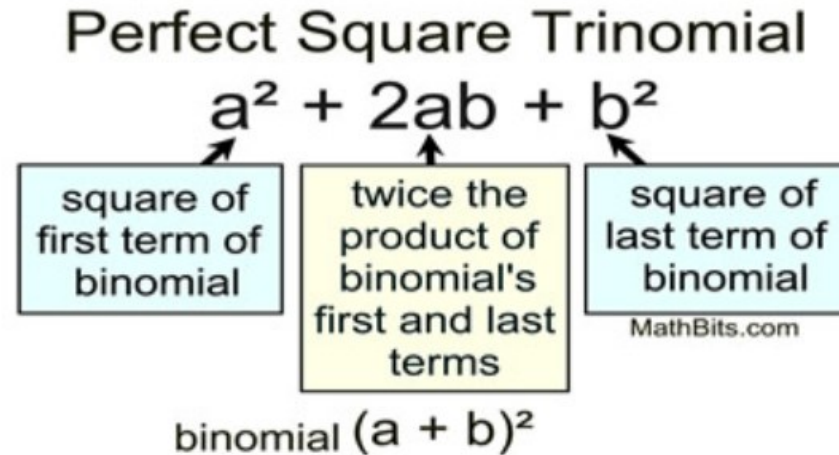
We say that $x - 2$ and $x + 2$ are **factors** of $x^2 - 4$.

Recognizing perfect squares

Squaring a binomial creates a perfect square trinomial:

$$(a + b)^2$$

$$(a - b)^2$$



Factor each trinomial.

(a) $x^2 + 6x + 9$

(b) $4x^2 - 4xy + y^2$

SOLUTION

- (a) Here $A = x$ and $B = 3$, so $2AB = 2 \cdot x \cdot 3 = 6x$. Since the middle term is $6x$, the trinomial is a perfect square. By the Perfect Square Formula we have

$$x^2 + 6x + 9 = (x + 3)^2$$

- (b) Here $A = 2x$ and $B = y$, so $2AB = 2 \cdot 2x \cdot y = 4xy$. Since the middle term is $-4xy$, the trinomial is a perfect square. By the Perfect Square Formula we have

$$4x^2 - 4xy + y^2 = (2x - y)^2$$

EXAMPLE

Factoring by grouping terms

Factor each polynomial.

(a) $x^3 + x^2 + 4x + 4$ (b) $x^3 - 2x^2 - 9x + 18$

SOLUTION

$$\begin{aligned}\text{(a)} \quad x^3 + x^2 + 4x + 4 &= (x^3 + x^2) + (4x + 4) \\ &= x^2(x + 1) + 4(x + 1) \\ &= (x^2 + 4)(x + 1) \\ \text{(b)} \quad x^3 - 2x^2 - 9x + 18 &= (x^3 - 2x^2) - (9x - 18) \\ &= x^2(x - 2) - 9(x - 2) \\ &= (x^2 - 9)(x - 2) \\ &= (x - 3)(x + 3)(x - 2)\end{aligned}$$

1.1 Algebraic Expressions: Find Sum, Difference, Product ¹⁸

(a) $(5 - 3x) + (2x - 8)$

(b) $(3x^2 + x + 1) - (2x^2 - 3x - 5)$

(c) $3(x - 1) + 4(x + 2)$

ACTIVITY 1.1

1.1 Algebraic Expressions: Find Sum, Difference, Product

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$$\begin{aligned}\text{(a)} \quad & (5 - 3x) + (2x - 8) \\ &= -x - 3\end{aligned}$$

$$\text{(b)} \quad (3x^2 + x + 1) - (2x^2 - 3x - 5)$$

Change the sign of the terms in the second parentheses and simplify by combining like terms:

$$\begin{aligned}&= 3x^2 + x + 1 - 2x^2 + 3x + 5 \\ &= x^2 + 4x + 6\end{aligned}$$

$$\text{(c)} \quad 3(x - 1) + 4(x + 2)$$

$$= 3x - 3 + 4x + 8 \quad \text{Expand the brackets}$$

$$= 7x + 5 \quad \text{Collect the like terms:}$$

SOLUTIONS

1.2 Algebraic Expressions: Using FOIL

(a) $(4s - 1)(2s + 5)$

(b) $(7y - 3)(2y - 1)$

(c) $(4x - 5y)(3x - y)$

ACTIVITY 1.2

1.2 Algebraic Expressions: Using FOIL

(a) $(4s - 1)(2s + 5)$

Use the FOIL method: First, Outside, Inside, Last.

$$\begin{array}{r} 8s^2 + 20s \quad -2s + -5 \\ 8s^2 + 18s - 5 \end{array}$$

(b) $(7y - 3)(2y - 1)$

$$\begin{array}{r} 14y^2 + -7y \quad -6y + 3 \\ 14y^2 - 13y + 3 \end{array}$$

(c) $(4x - 5y)(3x - y)$

$$\begin{array}{r} 12x^2 \quad -4xy \quad -15xy + 5y^2 \\ 12x^2 - 19xy + 5y^2 \end{array}$$

1.3 Algebraic Expressions: Multiplying

(a) $(x + 1)(2x^2 - x + 1)$

(b) $(1 + 2x)(x^2 - 3x + 1)$

(c) $x^{3/2}(\sqrt{x} - 1/\sqrt{x})$

ACTIVITY 1.3

5.3 Algebraic Expressions: Multiplying

$$(a) (x + 1)(2x^2 - x + 1)$$

$$2x^3 - x^2 + x + 2x^2 - x + 1 \quad \text{Multiply}$$

$$2x^3 + (2x^2 - x^2) + (x - x) + 1 \quad \text{Collect like terms}$$

$$2x^3 + x^2 + 1 \quad \text{Simplify}$$

$$(b) (1 + 2x)(x^2 - 3x + 1)$$

$$x^2 - 3x + 1 + 2x^3 - 6x^2 + 2x \quad \text{Multiply}$$

$$2x^3 + (x^2 - 6x^2) + (2x - 3x) + 1 \quad \text{Collect like terms}$$

$$2x^3 - 5x^2 - x + 1 \quad \text{Simplify}$$

$$(c) x^{3/2}(\sqrt{x} - 1/\sqrt{x}) \quad \text{Multiply}$$

$$x^{3/2}\left(x^{1/2} - \frac{1}{x^{1/2}}\right) \quad \text{Convert the fraction}$$

$$x^{3/2}(x^{1/2} - x^{-1/2}) \quad \text{Work out the exponents}$$

$$x^2 - x \quad \text{Simplify}$$

1.4 Algebraic Expressions: Factoring

(a) $x^2 - 14x + 48$

(b) $x^2 + 10x + 25$

(c) $\left(1 + \frac{1}{x}\right)^2 - \left(1 - \frac{1}{x}\right)^2$

ACTIVITY 1.4

1.4 Algebraic Expressions: Factoring

(a) $x^2 - 14x + 48$

The two factors of 48 that add to -14 are -6 and -8, therefore these terms are inserted into binomial expressions with x as the second term

$$(x - 6)(x - 8)$$

(b) $x^2 + 10x + 25$

Use the special factoring formula of perfect squares

$$\begin{aligned}(x)^2 + 2(5)(x) + (5)^2 \\ = (x + 5)^2\end{aligned}$$

1.4 Algebraic Expressions: Factoring

$$(c) \left(1 + \frac{1}{x}\right)^2 - \left(1 - \frac{1}{x}\right)^2$$

Factor the expression completely :

$$\left(1 + \frac{1}{x}\right)^2 - \left(1 - \frac{1}{x}\right)^2$$

Square of a Sum Formula: $(A + B)^2 = A^2 + 2AB + B^2$

Square of a Difference Formula: $(A - B)^2 = A^2 - 2AB + B^2$

Use the Square of a Sum and Square of a Difference Formulas for both terms respectively

$$\left(1 + \frac{1}{x}\right)^2 = 1^2 + (2 \times 1 \times \frac{1}{x}) + \left(\frac{1}{x}\right)^2$$

$$\left(1 - \frac{1}{x}\right)^2 = 1^2 - (2 \times 1 \times \frac{1}{x}) + \left(\frac{1}{x}\right)^2$$

$$= \left(1 + \frac{2}{x} + \frac{1}{x^2}\right) - \left(1 - \frac{2}{x} + \frac{1}{x^2}\right)$$

Distribute the minus sign (-1) to $\left(1 - \frac{2}{x} + \frac{1}{x^2}\right)$ and simplify

$$\left(1 + \frac{2}{x} + \frac{1}{x^2}\right) + \left(-1 + \frac{2}{x} - \frac{1}{x^2}\right)$$

$$= 1 + \frac{2}{x} + \frac{1}{x^2} - 1 + \frac{2}{x} - \frac{1}{x^2}$$

$$= \frac{4}{x}$$

Can you think of an easier way to solve this problem?

[Hint: think $a^2 - b^2$]

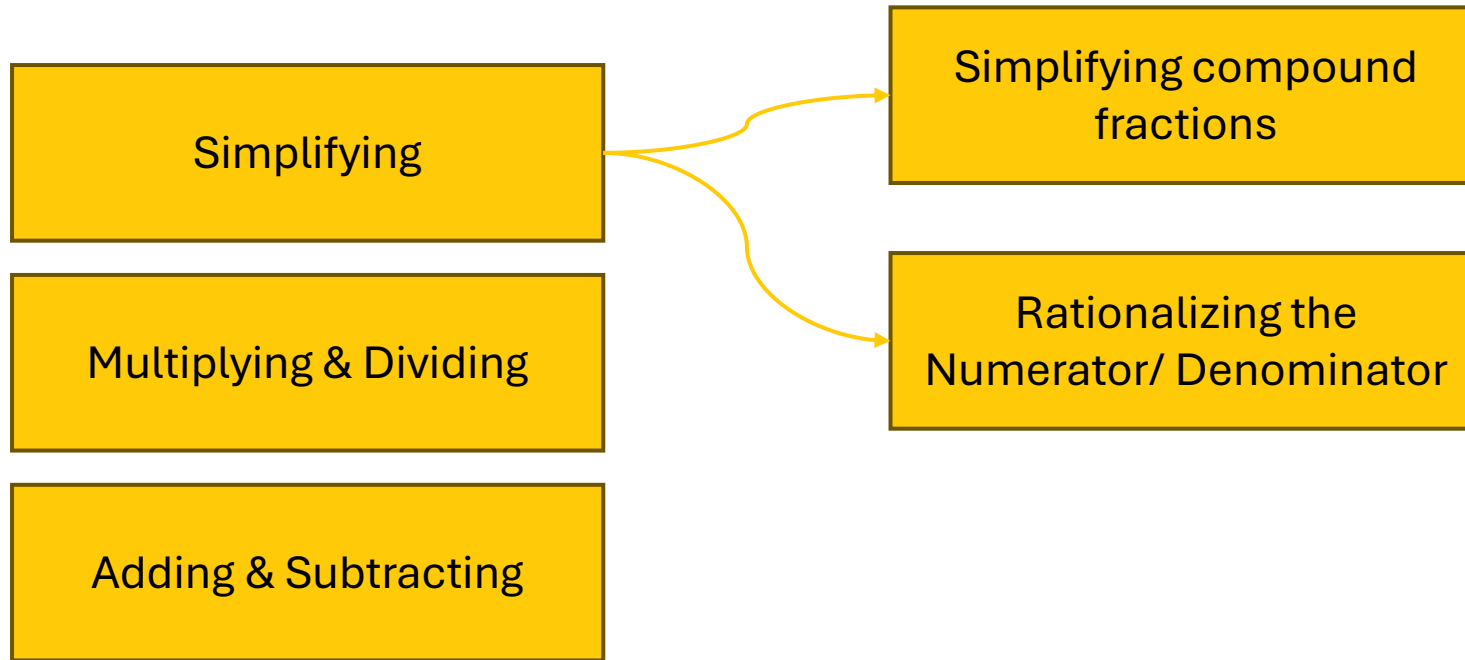


SOLUTIONS

2 Rational expressions

Rational expressions

Overview



Textbook Reference

Pre-calculus : Mathematics for Calculus
By Stewart, Redlin, Watson (7th Ed)

Chapter 1, Section 1.4:
Rational Expressions
Pg 36-42

What is a rational expression?

$$\underbrace{\frac{2x}{x-1} \quad \frac{y-2}{y^2+4} \quad \frac{x^3-x}{x^2-5x+6}}_{\text{Both numerator and denominator are polynomials}}$$

Both numerator and
denominator are
polynomials

$$\frac{x}{\underbrace{\sqrt{x^2+1}}_{\text{radical}}}$$

Not a rational expression
as denominator has a
radical

Simplifying rational expressions

$$\frac{x^3 + x^2}{x + 1}$$

simplifying often requires
factoring polynomials

Cancel common factors!

$$\frac{\cancel{AC}}{\cancel{BC}} = \frac{A}{B}$$

Simplify: $\frac{x^2 - 1}{x^2 + x - 2}$

SOLUTION

$$\begin{aligned} \frac{x^2 - 1}{x^2 + x - 2} &= \frac{(x - 1)(x + 1)}{(x - 1)(x + 2)} \\ &= \frac{x + 1}{x + 2} \end{aligned}$$

Factor

Cancel common factors

EXAMPLE

Multiplying and dividing rational expressions

To multiply rational expressions, we use the following property of fractions:

$$\frac{A}{B} \cdot \frac{C}{D} = \frac{AC}{BD}$$

Perform the indicated multiplication and simplify: $\frac{x^2 + 2x - 3}{x^2 + 8x + 16} \cdot \frac{3x + 12}{x - 1}$

SOLUTION We first factor.

$$\begin{aligned} \frac{x^2 + 2x - 3}{x^2 + 8x + 16} \cdot \frac{3x + 12}{x - 1} &= \frac{(x - 1)(x + 3)}{(x + 4)^2} \cdot \frac{3(x + 4)}{x - 1} && \text{Factor} \\ &= \frac{3(\cancel{x - 1})(x + 3)(\cancel{x + 4})}{(\cancel{x - 1})(x + 4)^2} && \text{Property of fractions} \\ &= \frac{3(x + 3)}{x + 4} && \text{Cancel common factors} \end{aligned}$$

EXAMPLE

Adding and Subtracting Rational expressions

To **add or subtract rational expressions**, we first find a common denominator and then use the following property of fractions:

$$\frac{A}{C} + \frac{B}{C} = \frac{A + B}{C}$$

Any denominator should work;

Often recommended to Use
the Least common
denominator (LCD)

(a) $\frac{3}{x-1} + \frac{x}{x+2}$

SOLUTION

(a) Here the LCD is simply the product $(x-1)(x+2)$.

$$\begin{aligned} \frac{3}{x-1} + \frac{x}{x+2} &= \frac{3(x+2)}{(x-1)(x+2)} + \frac{x(x-1)}{(x-1)(x+2)} && \text{Write fractions using LCD} \\ &= \frac{3x+6+x^2-x}{(x-1)(x+2)} && \text{Add fractions} \\ &= \frac{x^2+2x+6}{(x-1)(x+2)} && \text{Combine terms in numerator} \end{aligned}$$

EXAMPLE

Simplifying a compound fraction

Simplify: $\frac{\frac{x}{y} + 1}{1 - \frac{y}{x}}$

 Fractions in numerator and denominator!

SOLUTION 1 We combine the terms in the numerator into a single fraction. We do the same in the denominator. Then we invert and multiply.

$$\begin{aligned}\frac{\frac{x}{y} + 1}{1 - \frac{y}{x}} &= \frac{\frac{x+y}{y}}{\frac{x-y}{x}} = \frac{x+y}{y} \cdot \frac{x}{x-y} \\ &= \frac{x(x+y)}{y(x-y)}\end{aligned}$$

EXAMPLE

Rationalizing the denominator

Rationalize the denominator: $\frac{1}{1 + \sqrt{2}}$

SOLUTION We multiply both the numerator and the denominator by the conjugate radical of $1 + \sqrt{2}$, which is $1 - \sqrt{2}$.

$$\begin{aligned} \frac{1}{1 + \sqrt{2}} &= \frac{1}{1 + \sqrt{2}} \cdot \frac{1 - \sqrt{2}}{1 - \sqrt{2}} && \text{Multiply numerator and denominator by the conjugate radical} \\ &= \frac{1 - \sqrt{2}}{1^2 - (\sqrt{2})^2} && \text{Special Product Formula 1} \\ &= \frac{1 - \sqrt{2}}{1 - 2} = \frac{1 - \sqrt{2}}{-1} = \sqrt{2} - 1 \end{aligned}$$

EXAMPLE

Rationalizing the numerator

Rationalize the numerator: $\frac{\sqrt{4+h}-2}{h}$

SOLUTION We multiply numerator and denominator by the conjugate radical $\sqrt{4+h}+2$.

$$\frac{\sqrt{4+h}-2}{h} = \frac{\sqrt{4+h}-2}{h} \cdot \frac{\sqrt{4+h}+2}{\sqrt{4+h}+2}$$

Multiply numerator and denominator by the conjugate radical

$$= \frac{(\sqrt{4+h})^2 - 2^2}{h(\sqrt{4+h}+2)}$$

Special Product Formula 1

$$= \frac{4+h-4}{h(\sqrt{4+h}+2)}$$

$$= \frac{h}{h(\sqrt{4+h}+2)} = \frac{1}{\sqrt{4+h}+2}$$

Property 5 of fractions
(cancel common factors)

Conjugates and Conjugate radicals

$3x + 1$
 Conjugate: $3x - 1$

Expression		Its Conjugate
$x^2 - 3$	$\Leftrightarrow$	$x^2 + 3$
$a + b$	$\Leftrightarrow$	$a - b$
$a - b^3$	$\Leftrightarrow$	$a + b^3$

identical terms but **opposite arithmetic operators**
in the middle

Two binomials of the form $\sqrt{a} + \sqrt{b}$ and $\sqrt{a} - \sqrt{b}$ are called **conjugate radicals**:

$\sqrt{a} + \sqrt{b}$ is the conjugate of $\sqrt{a} - \sqrt{b}$

$\sqrt{a} - \sqrt{b}$ is the conjugate of $\sqrt{a} + \sqrt{b}$

For example, the conjugate of $2 - \sqrt{3}$ is $2 + \sqrt{3}$

Algebra: Common errors to avoid

Don't make the mistake of applying properties of multiplication to the operation of addition.
Many of the common errors in algebra involve doing just that.

Correct multiplication property	Common error with addition
$(a \cdot b)^2 = a^2 \cdot b^2$	$(a + b)^2 \neq a^2 + b^2$
$\sqrt{a \cdot b} = \sqrt{a} \sqrt{b} \quad (a, b \geq 0)$	$\sqrt{a + b} \neq \sqrt{a} + \sqrt{b}$
$\sqrt{a^2 \cdot b^2} = a \cdot b \quad (a, b \geq 0)$	$\sqrt{a^2 + b^2} \neq a + b$
$\frac{1}{a} \cdot \frac{1}{b} = \frac{1}{a \cdot b}$	$\frac{1}{a} + \frac{1}{b} \neq \frac{1}{a + b}$
$\frac{ab}{a} = b$	$\frac{a + b}{a} \neq b$
$a^{-1} \cdot b^{-1} = (a \cdot b)^{-1}$	$a^{-1} + b^{-1} \neq (a + b)^{-1}$

2.1 Simplify the following rational expressions

(a) $\frac{4(x^2-1)}{12(x+2)(x-1)}$

(b) $\frac{x^2-x-2}{x^2-1}$

(c) $\frac{x^2-25}{x^2-16} \cdot \frac{x+4}{x+5}$

(d) $\frac{x^2-x-6}{x^2+2x} \times \frac{x^3+x^2}{x^2-2x-3}$

(e) $\frac{3x-2}{x+1} - 2$

(f) $\frac{1}{x+1} + \frac{1}{x-1}$

ACTIVITY 2.1

2.1(a),(b) Simplify the following rational expressions

$$\begin{aligned}\text{(a)} \quad \frac{4(x^2-1)}{12(x+2)(x-1)} &= \frac{4(x-1)(x+1)}{12(x+2)(x-1)} \\ &= \frac{(x+1)}{3(x+2)}\end{aligned}$$

$$\begin{aligned}\text{(b)} \quad \frac{x^2-x-2}{x^2-1} &= \frac{(x-2)(x+1)}{(x+1)(x-1)} \\ &= \frac{x-2}{x-1}\end{aligned}$$

2.1(c) Simplify the following rational expressions

$$(c) \frac{x^2 - 25}{x^2 - 16} \cdot \frac{x + 4}{x + 5}$$

$$= \frac{x^2 - 25}{x^2 - 16} \times \frac{x + 4}{x + 5} \quad \text{Multiply the two fractions}$$

$$= \frac{(x^2 - 25)(x + 4)}{(x^2 - 16)(x + 5)} \quad \text{Factorise } x^2 - 25$$

$$= \frac{(x - 5)(x + 5)(x + 4)}{(x^2 - 16)(x + 5)} \quad \text{Factorise } x^2 - 16$$

$$= \frac{(x - 5)(x + 5)(x + 4)}{(x - 4)(x + 4)(x + 5)} \quad \text{Simplify the fraction}$$

$$= \frac{(x - 5)}{(x - 4)}$$

2.1(d),(e) Simplify the following rational expressions

$$(d) \quad \frac{x^2 - x - 6}{x^2 + 2x} \times \frac{x^3 + x^2}{x^2 - 2x - 3}$$

Factorise the fraction $\frac{x^2 - x - 6}{x^2 + 2x}$

$$= \frac{(x-3)(x+2)}{x(x+2)}$$

$$= \frac{x-3}{x}$$

Factorise the fraction $\frac{x^3 + x^2}{x^2 - 2x - 3}$

$$= \frac{x^2(x+1)}{(x-3)(x+1)}$$

$$= \frac{x^2}{(x-3)}$$

Multiply the two results:

$$= \frac{x-3}{x} \times \frac{x^2}{(x-3)}$$

$$= \frac{x^2(x-3)}{x(x-3)} \quad \text{Cancel out } x-3:$$

$$= x$$

$$(e) \quad \frac{3x-2}{x+1} - 2$$

$$= \frac{3x-2-(2x+2)}{x+1} \quad \text{Convert the term 2 to a fraction}$$

$$= \frac{3x-2}{x+1} - \frac{2}{1} \quad \text{Multiply the element by the denominator of the fraction containing variables}$$

$$= \frac{3x-2}{x+1} - \frac{2 \times (x+1)}{1 \times (x+1)} \quad \text{Expand the fraction}$$

$$= \frac{3x-2}{x+1} - \frac{2x+2}{x+1} \quad \text{Combine the fractions}$$

$$= \frac{3x-2-(2x+2)}{x+1}$$

$$= \frac{3x-2-2x-2}{x+1} \quad \text{Expand the negative sign}$$

$$= \frac{x-4}{x+1} \quad \text{Collect the like terms}$$

2.1(f) Simplify the following rational expressions

$$(f) \frac{1}{x+1} + \frac{1}{x-1}$$

$$= \frac{1 \times (x-1)}{(x-1)(x+1)} + \frac{1 \times (x+1)}{(x-1)(x+1)} \quad \text{Find the lowest common denominator}$$

$$= \frac{x-1}{(x-1)(x+1)} + \frac{x+1}{(x-1)(x+1)} \quad \text{Combine the fractions}$$

$$= \frac{x-1+x+1}{(x-1)(x+1)} \quad \text{Collect like terms}$$

$$= \frac{2x}{(x-1)(x+1)}$$

2.2 Compound fractions

$$(a) \quad \frac{1 - \frac{2}{y}}{\frac{3}{y} - 1}$$

$$(b) \quad \frac{x + \frac{y}{x}}{y + \frac{x}{y}}$$

ACTIVITY 2.2

2.2 (a) Compound fractions

$$(a) \quad \frac{1 - \frac{2}{y}}{\frac{3}{y} - 1}$$

Looking at the numerator:

Combine $1 - \frac{2}{y}$ by multiplying 1 by $\frac{y}{y}$. This then becomes: $\frac{y-2}{y}$.

Looking at the denominator:

Combine $\frac{3}{y} - 1$ by multiplying 1 by $\frac{y}{y}$. This then becomes: $\frac{3-y}{y}$.

The fraction is now

$$= \frac{\frac{y-2}{y}}{\frac{3-y}{y}}$$

Divide the fractions:

$$= \frac{(y-2)y}{y(3-y)}$$

Cancel the common factor of y :

$$= \frac{(y-2)}{(3-y)}$$

6.2 (b) Compound fractions

$$(b) \frac{x + \frac{y}{x}}{y + \frac{x}{y}}$$

Evaluate the sums:

$$\frac{x + \frac{y}{x}}{y + \frac{x}{y}} = \frac{\frac{x^2 + y}{x}}{\frac{y^2 + x}{y}}$$

Evaluate the division and simplify:

$$\dots = \frac{y(x^2 + y)}{x(y^2 + x)}$$

2.3 Rationalize the numerator/ denominator

(a) $\frac{1}{\sqrt{x} + 1}$

(b) $\frac{\sqrt{3} + \sqrt{5}}{2}$

ACTIVITY 2.3

2.3 (a) Rationalize the numerator/ denominator

$$(a) \frac{1}{\sqrt{x} + 1}$$

Multiply both numerator and denominator by $-\sqrt{x} + 1$ and simplify:

$$\frac{1}{\sqrt{x} + 1} = \left(\frac{1}{\sqrt{x} + 1} \right) \left(\frac{-\sqrt{x} + 1}{-\sqrt{x} + 1} \right)$$

$$= \frac{1 - \sqrt{x}}{1 - x}$$

$$= \frac{\sqrt{x} - 1}{x - 1}$$

2.3 (b) Rationalize the numerator/ denominator

$$(b) \quad \frac{\sqrt{3} + \sqrt{5}}{2}$$

Multiply both numerator and denominator by $\sqrt{3} - \sqrt{5}$ and simplify:

$$\begin{aligned} \frac{\sqrt{3} + \sqrt{5}}{2} &= \left(\frac{\sqrt{3} + \sqrt{5}}{2} \right) \left(\frac{\sqrt{3} - \sqrt{5}}{\sqrt{3} - \sqrt{5}} \right) \\ &= \frac{(\sqrt{3})^2 - (\sqrt{5})^2}{2\sqrt{3} - 2\sqrt{5}} \\ &= \frac{3 - 5}{2\sqrt{3} - 2\sqrt{5}} \\ &= -\frac{2}{2(\sqrt{3} - \sqrt{5})} \\ &= -\frac{1}{\sqrt{3} - \sqrt{5}} \end{aligned}$$

3. Equations

Equations

Overview

Solving
linear equations

Solving
**Quadratic
equations**

Solving equations
with
radicals and
Absolute values

Textbook Reference

Pre-calculus : Mathematics for Calculus
By Stewart, Redlin, Watson (7th Ed)

**Chapter 1, Section 1.5:
Equations**

Pg 45-55

What is an equation?

a statement that two mathematical expressions are **equal**

The diagram illustrates the components of the equation $2x - 5 = 13$. The equation is written in large purple font. The equals sign is red. Labels in pink text with lines pointing to the corresponding parts are as follows:

- Coefficient** points to the 2 in $2x$.
- Variable** points to the x in $2x$.
- Operator** points to the minus sign $-$.
- Constant** points to the 5 in 5 .
- Term** labels are placed below brackets under $2x$, 5 , and 13 .
- Expression** labels are placed below brackets under the entire left side ($2x - 5$) and the right side (13).

What is the solution of an equation?

The **values of the unknown** that **make the equation true** are called the **solutions** or **roots** of the equation.

The **process of finding the solutions** is called ***solving the equation***.

$x = 3$ is a solution of the equation
 $4x + 7 = 19$, because substituting
 $x = 3$ makes the equation true:

$$x = 3$$

$$4(3) + 7 = 19 \quad \checkmark$$

Review your past knowledge!

Remember how to simplify expressions?

Order of performing operations



PEMDAS

Parentheses (incl. absolute values, brackets, fraction lines, radicals)

Exponents (and roots)

Multiplication & Division

Addition & Subtraction

Mnemonic:

“Please Excuse My Dear Aunt Sally”

Properties of Equality

1. $A = B \Leftrightarrow A + C = B + C$

Adding the same quantity to both sides of an equation gives an equivalent equation.

2. $A = B \Leftrightarrow CA = CB \quad (C \neq 0)$

Multiplying both sides of an equation by the same nonzero quantity gives an equivalent equation.

We perform the
same operation on both sides
of an equation
when solving it.

Linear equations

first-degree equation

A **linear equation** in one variable is an equation equivalent to one of the form

$$ax + b = 0$$

where a and b are real numbers and x is the variable.

each term is either
a constant
 or **a nonzero multiple**
 of the variable

Linear equations

$$4x - 5 = 3$$

$$2x = \frac{1}{2}x - 7$$

$$x - 6 = \frac{x}{3}$$

Nonlinear equations

$$x^2 + 2x = 8$$

Not linear; contains the square of the variable

$$\sqrt{x} - 6x = 0$$

Not linear; contains the square root of the variable

$$\frac{3}{x} - 2x = 1$$

Not linear; contains the reciprocal of the variable

Solving a Linear Equation

Solve the equation $7x - 4 = 3x + 8$.

SOLUTION We solve this by changing it to an equivalent equation with all terms that have the variable x on one side and all constant terms on the other.

$$7x - 4 = 3x + 8 \quad \text{Given equation}$$

$$(7x - 4) + 4 = (3x + 8) + 4 \quad \text{Add 4}$$

$$7x = 3x + 12 \quad \text{Simplify}$$

$$7x - 3x = (3x + 12) - 3x \quad \text{Subtract } 3x$$

$$4x = 12 \quad \text{Simplify}$$

$$\frac{1}{4} \cdot 4x = \frac{1}{4} \cdot 12 \quad \text{Multiply by } \frac{1}{4}$$

$$x = 3 \quad \text{Simplify}$$

CHECK YOUR ANSWER

$$x = 3:$$

$$\begin{aligned} \text{LHS} &= 7(3) - 4 \\ &= 17 \end{aligned}$$

$$\begin{aligned} \text{RHS} &= 3(3) + 8 \\ &= 17 \end{aligned}$$

$$\text{LHS} = \text{RHS} \quad \checkmark$$

Solving Quadratic equations

QUADRATIC EQUATIONS

A **quadratic equation** is an equation of the form

$$ax^2 + bx + c = 0$$

where a , b , and c are real numbers with $a \neq 0$.

ZERO-PRODUCT PROPERTY

$$AB = 0 \quad \text{if and only if} \quad A = 0 \quad \text{or} \quad B = 0$$

if we can factor the left-hand side of a quadratic equation,
we can solve it by
setting each factor equal to 0

Find all real solutions of the equation $x^2 + 5x = 24$.

SOLUTION We must first rewrite the equation so that the right-hand side is 0.

$$x^2 + 5x = 24$$

$$x^2 + 5x - 24 = 0 \quad \text{Subtract 24}$$

$$(x - 3)(x + 8) = 0 \quad \text{Factor}$$

$$x - 3 = 0 \quad \text{or} \quad x + 8 = 0 \quad \text{Zero-Product Property}$$

$$x = 3 \quad \quad \quad x = -8 \quad \text{Solve}$$

The solutions are $x = 3$ and $x = -8$.

EXAMPLE

Key concept: Completing the square

Completing the square of an expression with multiple variables is a technique which manipulates the expression into a **perfect square plus some constant.**

$$x^2 + 6x = 2$$

$$x^2 + 6x + 9 = 2 + 9$$

$$x^2 + 6x + 9 = 11$$

$$(x + 3)^2 = 11$$

This expression cannot be factored into a perfect square

We want to bring the polynomial to the form $(a + b)^2$

Half the **coefficient** of the term with degree 1 and **square** it;

$$6/2=3$$

$$3^2=9$$

Add this constant to both sides of equation

Completing the square: Formal definition

COMPLETING THE SQUARE

To make $x^2 + bx$ a perfect square, add $\left(\frac{b}{2}\right)^2$, the square of half the coefficient of x . This gives the perfect square

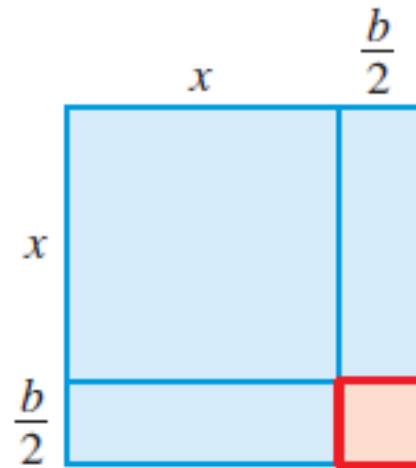
$$x^2 + bx + \left(\frac{b}{2}\right)^2 = \left(x + \frac{b}{2}\right)^2$$

Completing the Square

The area of the blue region is

$$x^2 + 2\left(\frac{b}{2}\right)x = x^2 + bx$$

Add a small square of area $(b/2)^2$ to “complete” the square.



An equation involving a radical

Solve the equation $2x = 1 - \sqrt{2 - x}$.

SOLUTION To eliminate the square root, we first isolate it on one side of the equal sign, then square.

$2x - 1 = -\sqrt{2 - x}$	Subtract 1
$(2x - 1)^2 = 2 - x$	Square each side
$4x^2 - 4x + 1 = 2 - x$	Expand LHS
$4x^2 - 3x - 1 = 0$	Add $-2 + x$
$(4x + 1)(x - 1) = 0$	Factor
$4x + 1 = 0$ or $x - 1 = 0$	Zero-Product Property
$x = -\frac{1}{4}$ $x = 1$	Solve

CHECK YOUR ANSWERS

$x = -\frac{1}{4}$:

LHS = $2(-\frac{1}{4}) = -\frac{1}{2}$

RHS = $1 - \sqrt{2 - (-\frac{1}{4})}$
 $= 1 - \sqrt{\frac{9}{4}}$
 $= 1 - \frac{3}{2} = -\frac{1}{2}$

LHS = RHS ✓

$x = 1$:

LHS = $2(1) = 2$

RHS = $1 - \sqrt{2 - 1}$
 $= 1 - 1 = 0$

LHS $\neq$ RHS ✗

EXAMPLE

An absolute value equation

When solving an absolute value equation, we use the following property

$$|X| = C \text{ is equivalent to } X = C \text{ or } X = -C$$

where X is any algebraic expression. This property says that to solve an absolute value equation, we must solve **two separate equations**.

Solve the equation $|2x - 5| = 3$.

SOLUTION By the definition of absolute value, $|2x - 5| = 3$ is equivalent to

$$2x - 5 = 3 \quad \text{or} \quad 2x - 5 = -3$$

$$2x = 8$$

$$2x = 2$$

$$x = 4$$

$$x = 1$$

The solutions are $x = 1, x = 4$.

3.1 Linear equations

(a) $3x + 4 = 7$

(b) $3 + \frac{1}{3}x = 5$

(c) $2x + 3 = 7 - 3x$

ACTIVITY 3.1

3.1 Linear equations

(a) $3x + 4 = 7$

Subtract 4 from both sides: $3x = 3$

Divide both sides by 3: $x = 1$

(b) $3 + \frac{1}{3}x = 5$

Subtract 3 from each side: $\frac{1}{3}x = 2$

Divide each side by $\frac{1}{3}$ (equivalent to multiplying each side by 3): $x = \frac{2}{\frac{1}{3}} = 6$

(c) $2x + 3 = 7 - 3x$

Add $3x$ to both sides: $5x + 3 = 7$

Subtract 3 from both sides: $5x = 4$

Divide both sides by 5: $x = \frac{4}{5}$

3.2 Solving by factoring

(a) $x^2 + 3x - 4 = 0$

(b) $2x^2 + 7x + 6 = 0$

ACTIVITY 3.2

3.2 Solving by factoring

(a) $x^2 + 3x - 4 = 0$

1. Factor the equation: The factors are $(x-1)$ and $(x+4)$ because $-1 \times 4 = -4$ (the c value in $a^2 + bx + c$ and $-1+4=3$, the b value).
2. Set the factors equal to zero: $x - 1 = 0$ and $x + 4 = 0$
3. Solve: $x = 1, -4$

(b) $2x^2 + 7x + 6 = 0$

$$2x^2 + 4x + 3x + 6 = 0$$

$$2x(x + 2) + 3(x + 2) = 0$$

$$(2x + 3)(x + 2) = 0$$

$$x = -\frac{3}{2} \quad x = -2$$

3.3 Completing the square

(a) $x^2 - 4x + 2 = 0$

(b) $x^2 + 3x - \frac{7}{4} = 0$

ACTIVITY 3.3

3.3a Completing the square

Let's take the independent term to the right side of the equation:

$$x^2 - 4x = -2$$

Let's remember, that in order to complete the square, we need to add $(\frac{b}{2})^2$ to both sides of the equation. b is the coefficient of the first degree term. In this case $b = -4$

$$x^2 - 4x + (\frac{-4}{2})^2 = -2 + (\frac{-4}{2})^2$$

$$x^2 - 4x + 4 = -2 + 4$$

$$x^2 - 4x + 4 = 2$$

We have a perfect square trinomial on the left side of the equation. We factor it and the equation becomes:

$$(x - 2)^2 = 2$$

Take the square root of both sides of the equation:

$$\sqrt{(x - 2)^2} = \sqrt{2}$$

$$x - 2 = \pm\sqrt{2}$$

Solve for x :

$$x = 2 \pm \sqrt{2}$$

3.3b Completing the square

$$(b) \quad x^2 + 3x - \frac{7}{4} = 0$$

$$x^2 + 3x - \frac{7}{4} = 0$$

Let's take the independent term to the right side of the equation:

$$x^2 + 3x = \frac{7}{4}$$

Remember that, in order to complete the square, we need to add $(\frac{b}{2})^2$ to both sides of the equation. b is always the coefficient of the first degree term. In this equation, $b = 3$:

$$x^2 + 3x + (\frac{3}{2})^2 = \frac{7}{4} + (\frac{3}{2})^2$$

$$x^2 + 3x + \frac{9}{4} = \frac{7}{4} + \frac{9}{4}$$

$$x^2 + 3x + \frac{9}{4} = \frac{16}{4}$$

$$x^2 + 3x + \frac{9}{4} = 4$$

We have a perfect square trinomial on the left side of the equation. We factor it and the equation becomes:

$$(x + \frac{3}{2})^2 = 4$$

Take the square root of both terms:

$$\sqrt{(x + \frac{3}{2})^2} = \sqrt{4}$$

$$x + \frac{3}{2} = \pm 2$$

Solve for x :

$$x = -\frac{3}{2} \pm 2$$

The two solutions are:

$$x = -\frac{3}{2} + 2 = \frac{1}{2} \text{ and } x = -\frac{3}{2} - 2 = -\frac{7}{2}$$

Solving equations with radicals and absolute values

(a) $\sqrt{8x - 1} = 3$

(b) $\sqrt{3 + x} = \sqrt{x^2 + 1}$

(c) $|2x| = 3$

(d) $|x - 6| = -1$

ACTIVITY 3.4

3.4 a, b: Solving equations with radicals

$$(a) \quad \sqrt{8x-1} = 3$$

square both sides of the equation

$$(\sqrt{8x-1})^2 = 3^2$$

$$8x - 1 = 9$$

$$8x = 10$$

Solve for x

$$x = \frac{10}{8} = \frac{5}{4}$$

$$(b) \quad \sqrt{3+x} = \sqrt{x^2+1}$$

square both sides of the equation

$$(\sqrt{3+x})^2 = (\sqrt{x^2+1})^2$$

Take all terms to the right side of the equation

$$3 + x = x^2 + 1$$

$$x^2 - x + 1 - 3 = 0$$

$$x^2 - x - 2 = 0$$

Solve by factoring

$$(x-2)(x+1) = 0$$

$$x = 2 \text{ and } x = -1$$

We get two solutions

SOLUTIONS

3.4 c, d: Solving equations with absolute values

(c)

$$|2x| = 3$$

By the definition of absolute value, solving this equation is equivalent to solving two separate equations, which are $2x = 3$ and $2x = -3$

Solve both equations for x :

$$\begin{aligned} 2x &= 3 \\ x &= \frac{3}{2} \end{aligned}$$

$$\begin{aligned} 2x &= -3 \\ x &= -\frac{3}{2} \end{aligned}$$

The two solutions are $x = \frac{3}{2}$ and $x = -\frac{3}{2}$

(d) $|x - 6| = -1$

There are no values of x that make this equation true, because an absolute value always yields a positive number.

This equation has no solution!

SOLUTIONS